

Processing and Simulation of Radiometric Observations Using the Open-Source TU Delft Astrodynamics Toolbox (Tudat)

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Background

Archival of radiometric observations in various formats:

- ODF, ATDF, and TNF (binary)
- IFMS, and CCSDS TDM (text)

Recent extension of Tudat to process various tracking data uniformly [Fig. 1]:

- For each data source:
 - Parse raw representation
 - Normalize into primitive container
- Convert primitive data containers into Tudat-linked object
 - Unified data structure, compatible with all observable types

This enables:

- Grouping of common functionality
- Future extension with minimal effort
- Integration with database systems such as GOOD-WISE [2].

Tracking Data Interface

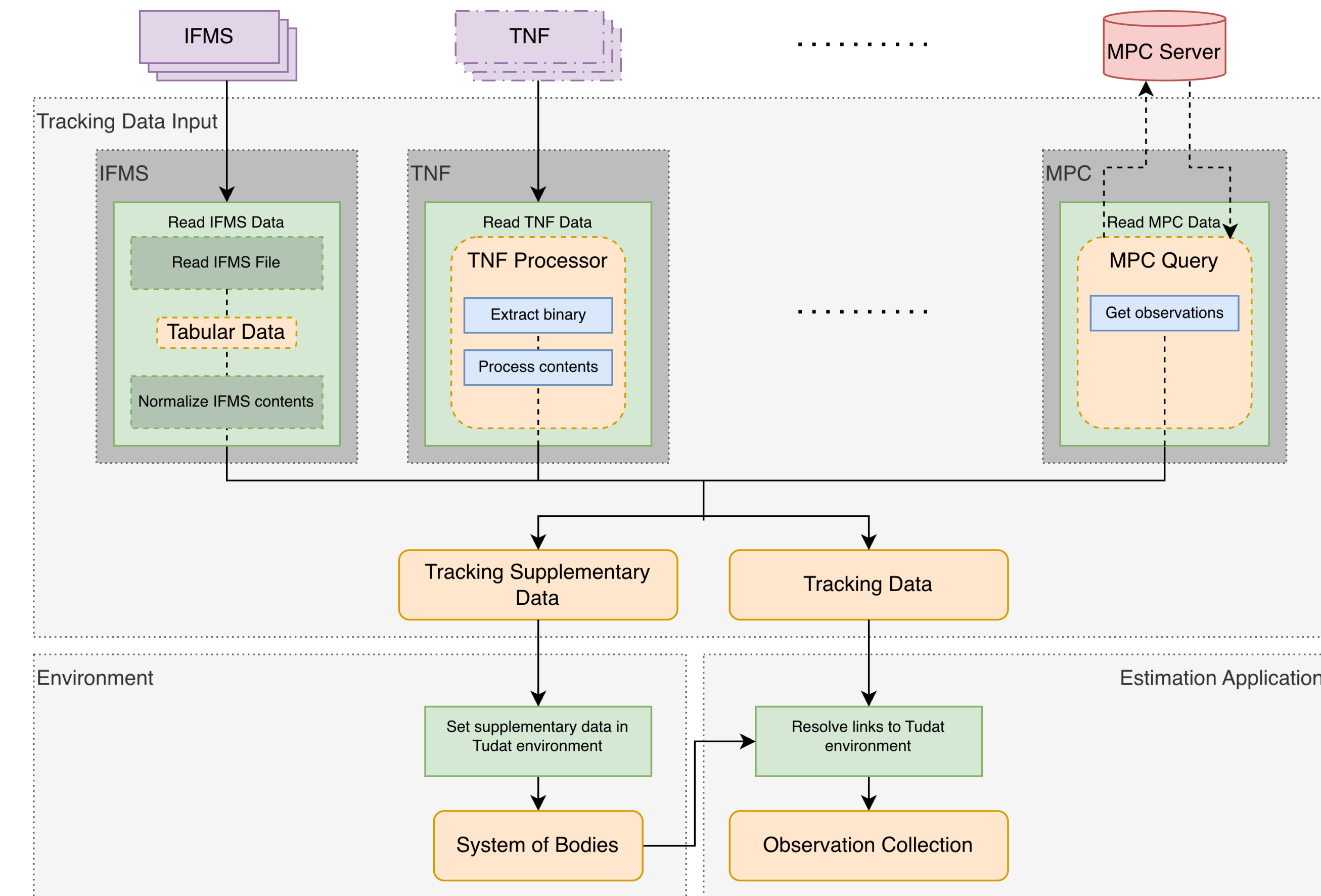


Fig. 1: Schematic of how tracking data is loaded into Tudat. Note that, while not discussed here further, this architecture is common for all observable types and not limited to radiometric observables.

Simulation Setup

Validation of tracking data interface and closed-loop observation models using **pre-fit analysis**:

- Compute observations along **reference trajectory**
- Compare **real observations** to **computed observations** to obtain pre-fit **residuals**

Support for the following file formats with example missions:

- A. ATDF files (using atdf2ascii [3]) Magellan
- B. ODF files GRAIL-A
- C. IFMS files Mars Express
- D. TNF files (using trk234 [4]) MRO

Computation of observables according to [5], including:

- Second-order relativistic effects: A, B, C, D
- Tropospheric effects:
 - TRK2-23 media correction [6]: A, B, D
 - Saastamoinen tropospheric model [7]: C
- Ionospheric effects:
 - TRK2-23 media correction [6]: A, B, D
- Solar corona effects [8]: C, D
- Observable simulation w.r.t. antenna center: B, C, D

Pre-fit Results

- Retrieval of **one month of closed-loop tracking data** observations per mission
- Compression to a standard **count-time of 60s**
- Filtering for gross outliers (>100mHz for Magellan, >20mHz for all other cases)

- Residuals are largely unbiased, with a root-mean square (RMS) in the order of **mHz** for **GRAIL, MEX and MRO**, and **10s of mHz** for **Magellan**
- Individual passes retain small signatures, consistent with a pre-fit analysis

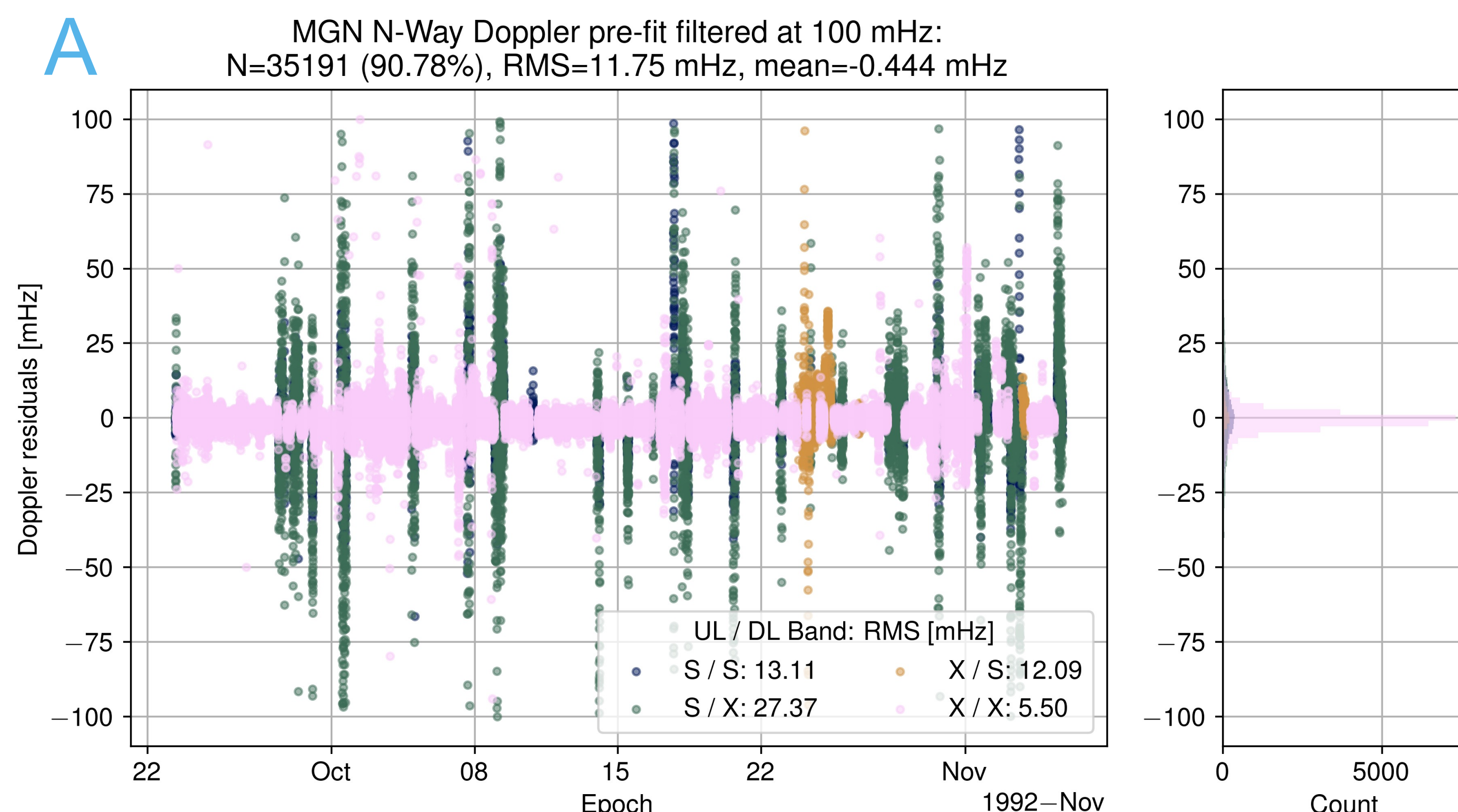


Fig. 2: Pre-fit Doppler residuals of Magellan, extracted from ATDF files using the atdf2ascii tool [3]. The observations are computed using the JPL GRAV orbit for mapping cycle 4 and are colored by the up- and downlink band.

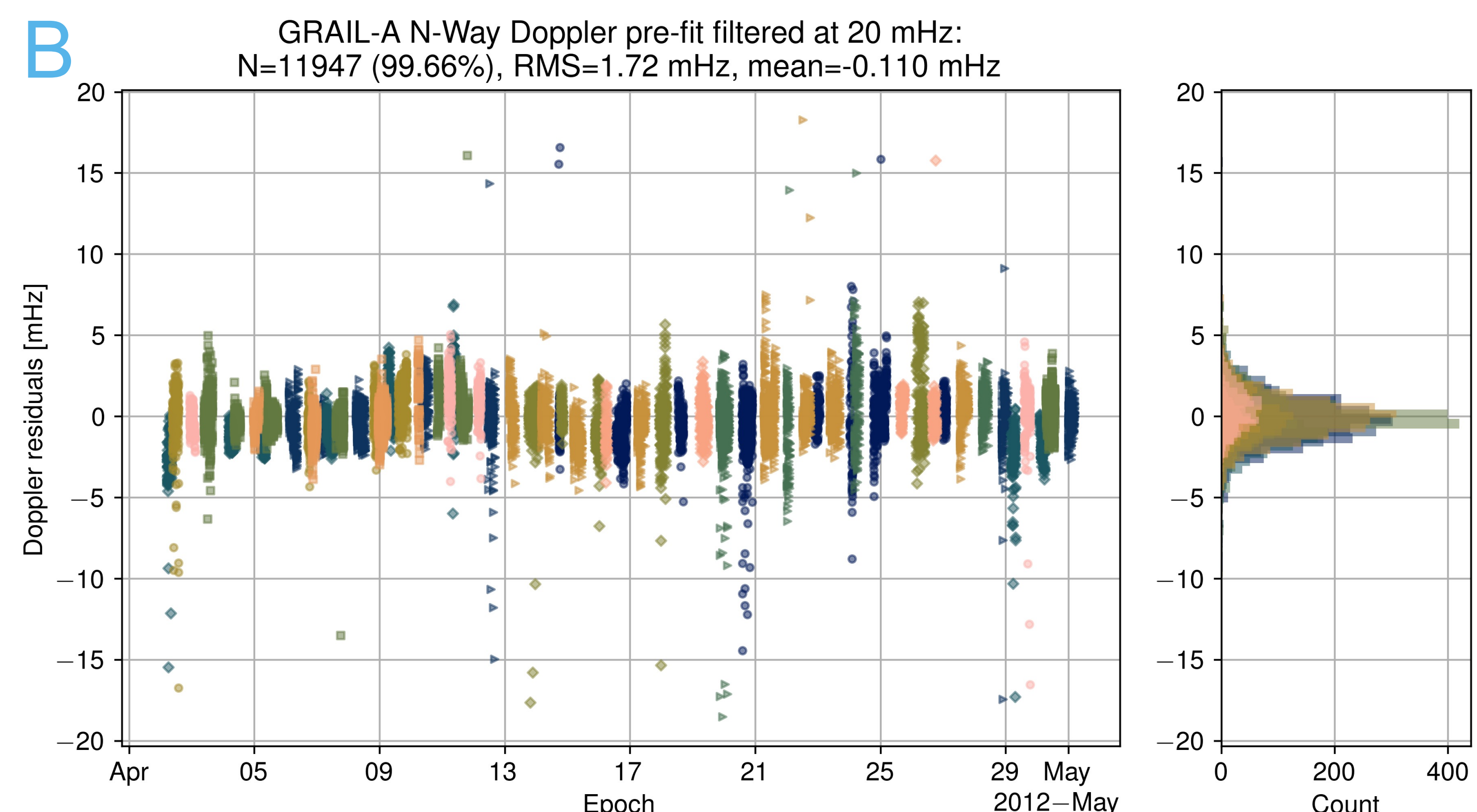


Fig. 3: Pre-fit Doppler residuals of GRAIL-A, extracted from ODF files. The observations are computed using the GRAIL Science/JPL orbit and are colored by link configuration (see the online version for the labels).

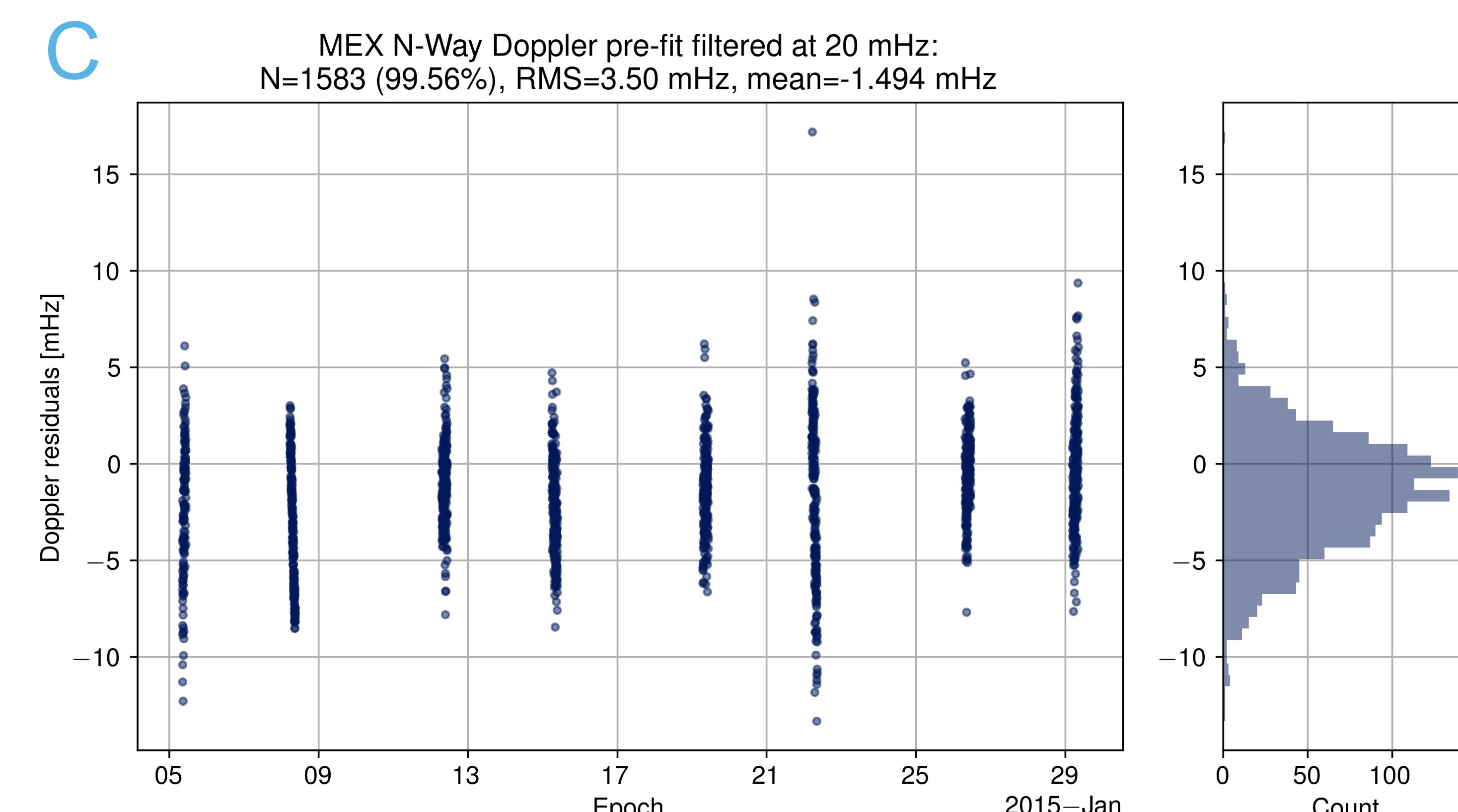


Fig. 4: Pre-fit Doppler residuals of Mars Express (MEX), extracted from IFMS Level 2 files taken from New Norcia in X-band. The observations are computed using the MEX Project Science team orbit.

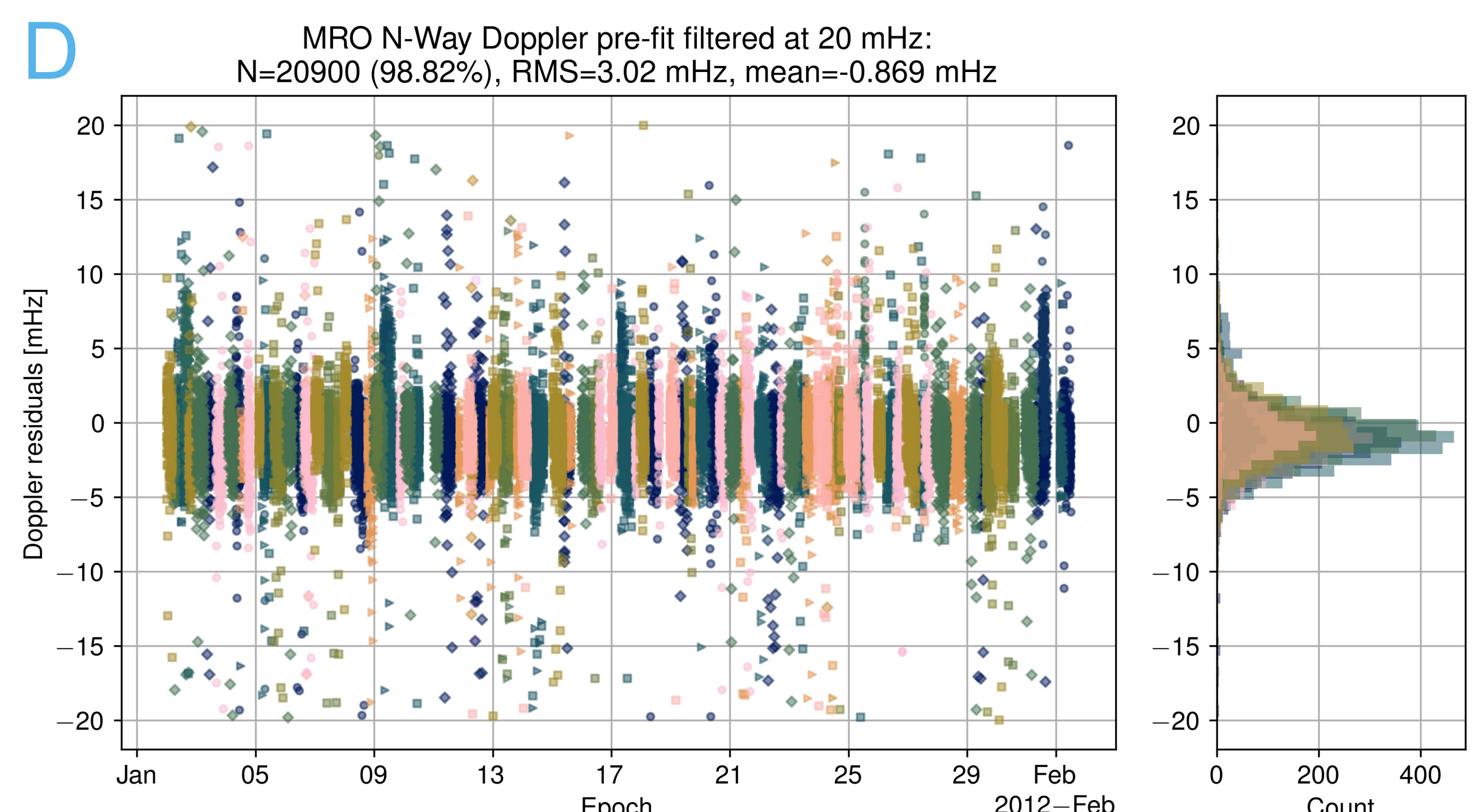


Fig. 5: Pre-fit Doppler residuals of MRO, extracted from TNF files using the trk234 tool [4]. The observations are computed using the MRO NAV team orbit and are colored by link configuration (see the online version for the labels).

Conclusions

- **Unified interface** to ingest **closed-loop radiometric** data, supporting current and historic formats
- Validation of pipeline for selected inner Solar system missions
- **Doppler pre-fits @ 60s** in the order of
 - **several mHz** (GRAIL, MEX, MRO)
 - **10s of mHz** (Magellan)

Future Work

- Add CCSDS TDM interface
- **Extend validation of full estimation pipeline** against reference solutions
- Apply estimation framework to **multi-mission, multi-observable type estimation** scenarios
- **Integrate Tudat with GOOD-WISE** for end-to-end reproducible estimation workflows
 - See [2] on Friday, 08:45, Room Earth (Tango 1)

References

- [1] Gisolfi et al. IAC. 2025
- [2] Dirkx et al. EPSC. 2026
- [3] Verma. SoftwareX. 2022
- [4] NASA Planetary Data System Software. PyTrk234. 2024
- [5] Moyer. Wiley. 2003
- [6] Machuzak. JPL Publication. 2008
- [7] Estefan et al. JPL Publication. 1994
- [8] Verma et al. Astronomy & Astrophysics. 2013.

Datasets used in this work (see repository for details)

Magellan: 10.17189/1522511
GRAIL-A: 10.17189/1519558
Mars Express: 10.57780/esa-53ktg7l, 10.57780/esa-gxidkt, 10.57780/esa-34ewi6a, 10.57780/esa-pgxj5wd, 10.57780/esa-30gu8gp, 10.57780/esa-h8yi287, 10.57780/esa-xwvg83d, 10.57780/esa-w8zdkpj
Mars Reconnaissance Orbiter: 10.17189/1519443

Links

Repository with analysis source code



TU Delft Astrodynamics Toolbox (Tudat)

